25901539 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSESDV200 - ELECTRO ACOUSTICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer

Answer any **8** questions

Each question carries **2** marks

1. What is the symbol used for AC supply? [U] / [CO1]
2. What is the color code for left & right channel in RCA standard? [U] / [CO2]
3. What causes the proximity effect in cardioid microphones? [U] / [CO3]
4. Define on-ear headphones. [U] / [CO4]
5. What is a boundary microphone? [U] / [CO3]
6. What is a rectifier? [U] / [CO1]
7. State one factor that affects frequency response in cables? [U] / [CO2]
8. State the unit used to express microphone sensitivity. [U] / [CO3]

9.	State one advantage of horn type loudspeakers.	[U] / [CO4]
10.	What is common mode rejection ratio (CMRR)?	[U] / [CO1]

[2x8 = 16]

Part B

Short answer

Answer any **5** questions

Each question carries **6** marks

11.	Explain the concept of stereophonic microphone systems.	[U] / [CO3]
12.	Explain the importance of frequency division in loudspeaker systems.	[U] / [CO4]
13.	Differentiate between line cable and speaker cable?	[U] / [CO2]
14.	Explain clipping distortion with a neat waveform diagram. How can it be minimized?	[U] / [CO2]
15.	Explain the working of a variable capacitance (condenser) microphone, highlighting the role of the diaphragm and backplate.	[U] / [CO3]
16.	Discuss the concept of loudspeaker directivity.	[U] / [CO4]
17.	Explain the relationship between voltage, current, and resistance ?	[U] / [CO1]

[6x5 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **12** marks

18.	Discuss the advantages and limitations of OFC in audio communication?	[U] / [CO2]
19.	Explain highly directional microphones in detail.	[U] / [CO3]
20.	Explain in detail the concept of loudspeaker baffles and enclosures.	[U] / [CO4]

[12x2 = 24]